

CODSOFT ARTIFICIAL INTELLIGENCE INTERNSHIP

Task 3: Recommendation System

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Project: Recommendation System Using Python

Introduction

This project is a python-based Recommendation system Developed as part of the CODSOFT Artificial Intelligence Internship Program.

The system suggest items such as movies, books, products, and music based on user preference. It demonstrates the basic concept of content-based filtering using python programing.

Objective

The main objective of this project:

- To develop a simple recommendation system using Python.
- To understand the concept of content-based filtering.
- To improve programing logic and problem-solving skills.
- To handle user input and display dynamic recommendations

Features

- Movies recommendations
- Books recommendation
- Products recommendation
- Music recommendation
- Simple command-line interface
- Continuous user interaction system

System Required

- **Hardware Requirements:**
 - RAM: Minimum 2GB
 - Storage: 500MB free space
 - Processor: Intel i3 or above
 - Computer/Laptop
- **Software Requirements:**
 - Python 3.x
 - Vs code or any python IDE
 - Command prompt/Terminal

Technology Used

- Python- Used for implementing Recommendation logic
- Vs Code- Used as Code editor and execution environment

Algorithm

- **Step 1:**
Start the program
- **Step2:**
Create a dictionary of categories and items
- **Step 3:**
Display available categories to the user
- **Step 4:**
Take input from user
- **Step 5:**
Check if input matches available categories
- **Step 6:**
If matched, display recommended items
- **Step 7:**
If not matched, show error message
- **Step 8:**
Repeat process until user enters "exit".
- **Step 9:**
Stop the program

Source code

```
print("=== Recommendation System ===")

items = {
    "movies": ["3 Idiots", "Dangal", "Taare Zameen Par", "12th
Fail"],
    "books" : ["Rich Dad Poor Dad", "The School of Life", "Atomic
Habits"],
    "products" : ["Laptops", "Headphones", "Smartphones"],
    "music" : ["Arijit Singh", "Lofi beats", "Anuv Jain"],
}

def recommend(category) :
    if category in items:
        print("\nRecommend",category, ":" )
        for i in items[category]:
            print("-", i)
    else:
        print("No recommendations found for this category.")

while True:
    print("\nCategories: movies / books / products / music /
exit")
    user = input("Enter your preference: ").lower()

    if user == "exit":
        print("Thank you for using recommendation system")
        break

    recommend(user)
```

Sample Output

Categories: movies / books / products / music / exit

Enter your preference: movies

Recommended movies:

- 3 Idiots
- Dangal
- Taare Zameen Par
- 12th Fail

The system displays:

- Recommended items based on category
- Error message for invalid input
- Exit message when user stops program

Learning Outcomes

Through this project I have learned:

- Basics of python programming
- Working with dictionaries and loops
- User input handling
- Logic building for recommendation system
- Understanding for content-based filtering

Future Enhancements

- Implement Machine Learning-based recommendation system
- Add user rating system
- User-based personalized recommendation system
- Mobile application version of the system

Conclusion

The Recommendation system project successfully demonstrates how python can be used to create a simple intelligent suggestion system. It improves logical thinking and programming skills and can be further enhanced using advanced AI techniques.

References

- Python Official Documentation
- Online Learning Resources
- CODSOFT Internship Guideline

Result

The Recommendation system was successfully developed and executed using python. It provides category-based recommendation and works efficiently in a command-line interface.